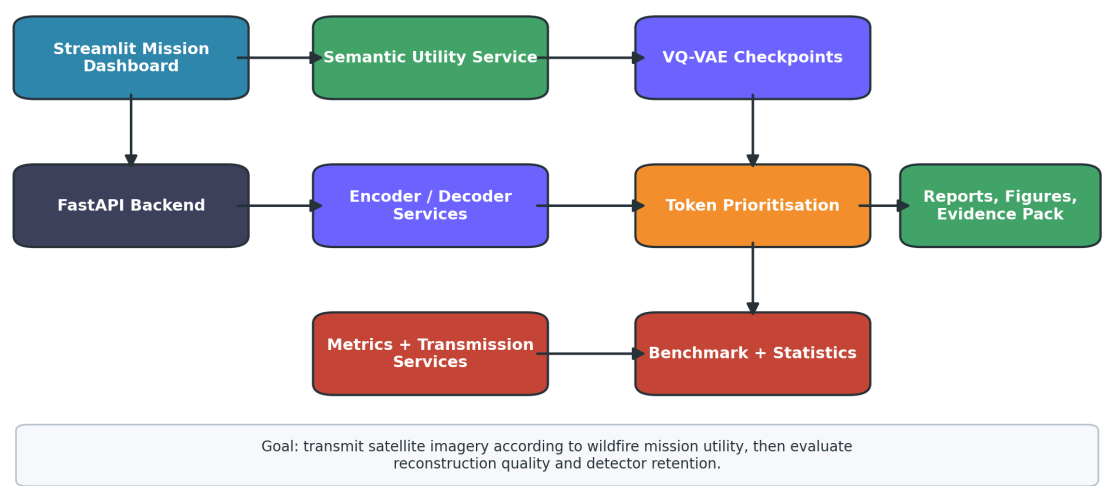


Founder and Visa Evidence Package

Onboard AI Software for Mission-Aware Satellite Downlink Optimisation

CompressAI System Architecture



Purpose: frame the same validated research platform for Innovator Founder, Global Talent, incubator, and supervisor-facing evidence.

1. Business Proposition

The proposed product is an onboard AI software layer that reduces satellite downlink bandwidth while preserving mission-critical Earth Observation information. The initial application is wildfire monitoring, where preserving fire, smoke, burn-scar, and affected-terrain evidence matters more than visually perfect reconstruction of every background pixel.

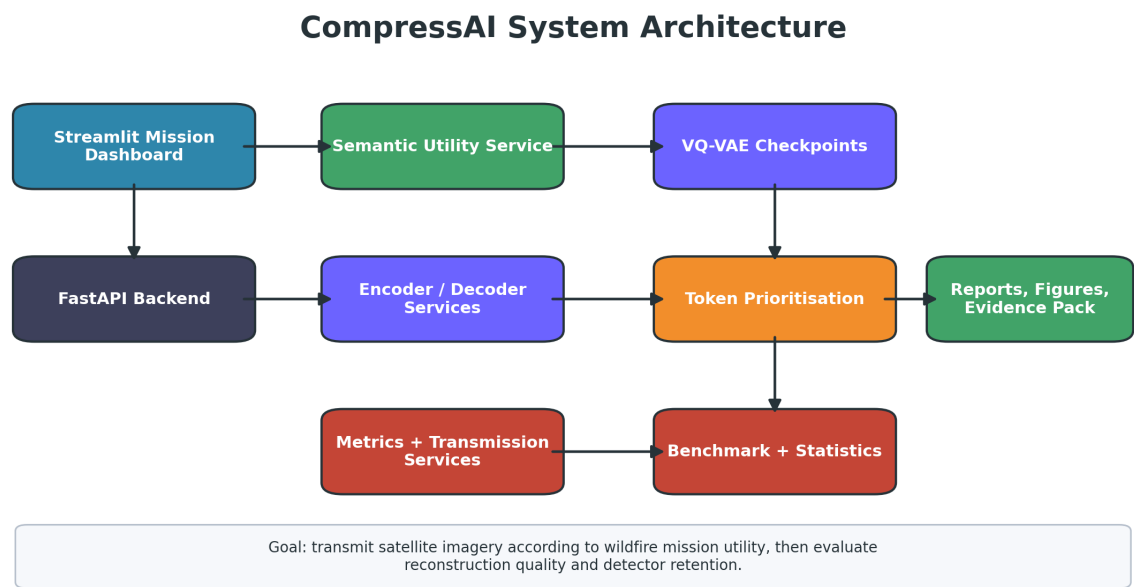
One-line positioning

Mission-aware semantic communication software for resource-constrained Earth Observation satellites.

2. Innovative, Viable, Scalable

Criterion	Evidence
Innovative	Utility-aware learned token transmission for satellite imagery rather than generic pixel compression.
Viable	Working API, dashboard, benchmark framework, Sentinel-2 evidence, statistical reports, and reproducible diagrams.
Scalable	Applicable to CubeSats, EO analytics, wildfire monitoring, flood validation, maritime monitoring, and defence/civil protection use cases.

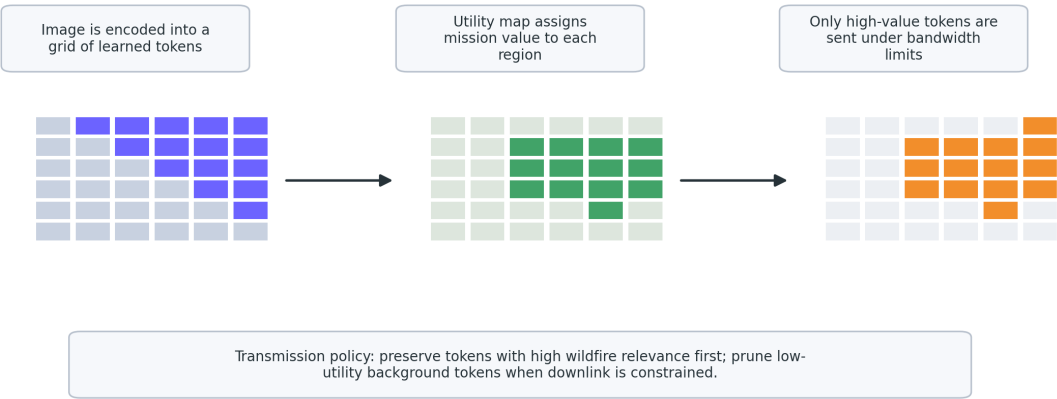
3. Platform Architecture



4. Current Technical Evidence

Metric	Original VQ-VAE	Regularized mixed VQ-VAE	Interpretation
SUS	80.470	81.159	Candidate slightly higher, but not enough alone for promotion
Detector retention	0.8642	0.8403	Original safer for mission-utility default
PSNR	21.305	22.631	Candidate clearly improves visual reconstruction
SSIM	0.8170	0.8354	Candidate clearly improves structural similarity
Bandwidth saved	98.56%	98.53%	Both demonstrate strong communication savings

Token-Based Semantic Transmission

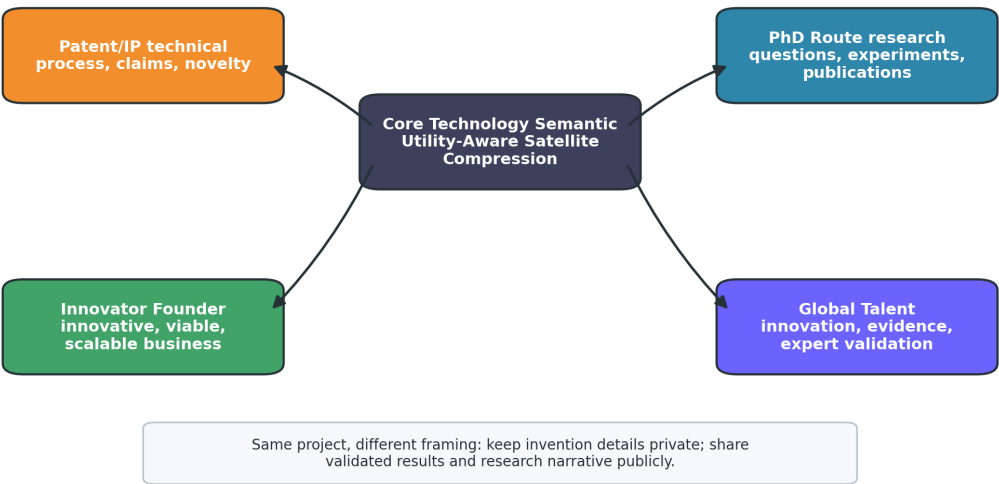


5. Global Talent Evidence Mapping

Evidence type	Current status	Next action
Original technical work	Working semantic compression platform	Prepare short public technical video
Research output	Reports, figures, Sentinel-2 validation	Submit arXiv or workshop preprint
Open-source contribution	GitHub repository available	Polish README and examples
External validation	Not yet enough	Collect supervisor and incubator feedback
Commercial promise	Founder business case drafted	Contact incubators and collect letters of interest

6. Career Evidence Positioning

Evidence Positioning: Patent, PhD, Innovator Founder, Global Talent



7. Immediate 60-Day Plan

- Create a 2 to 3 minute demo video showing image input, utility map, token selection, reconstruction, and metrics.
- Send the PhD package to 20 targeted supervisors in EO, semantic communication, onboard AI, and neural compression.
- Approach ESA BIC, Satellite Applications Catapult, and space-tech incubators with the founder package.
- Book patent-attorney consultation before publishing claim-level technical mechanisms.
- Prepare one arXiv or workshop-style preprint around mission-utility-aware semantic compression.

This package is evidence framing, not immigration advice. It is designed to support discussions with endorsing bodies, supervisors, incubators, and legal professionals.